Unbalanced Class Learning Network With Scale-Adaptive Perception for Complicated Scene in Remote Sensing Images Segmentation

He Wang, Mengmeng Zhang, Wei Li, *Senior Member, IEEE*, Yunhao Gao, Yuanyuan Gui, and Yuxiang Zhang, *Student Member, IEEE*

***Abstract*— The semantic segmentation of wide-field remote sensing images (RSIs) plays a significant role in many fields. However, due to the complexity of the content of RSIs, the dataset often has an uneven distribution of land type between different classes and large gaps in the scales of different objects. This often creates great problems for fine segmentation. To solve the issues, an unbalanced class learning network with scale-adaptive perception (UCSANet) is proposed, which can adaptively cope with multiscale objects and unbalanced classes. The design can be inserted in any convolution network easily and can enrich features without increasing too many parameters. The network groups feature and use atrous convolutions with different dilated rates on different groups to extract multiscale features while separable convolutions reduce the amount of network parameters. Then, the fusion of features between different scales is achieved through the self-attention mechanism. Furthermore, a weight map is designed to adaptively combine the predictions of two segmentation heads with cross-entropy loss and Lovasz-Softmax loss, respectively, which enable the network to focus on learning low-frequency classes without affecting high-frequency classes. Experimental results on GF-6 MSI datasets demonstrate that the proposed UCSANet performs significantly better than others and achieves multiclass segmentation more accurately.**

***Index Terms*— Deep learning, scale-adaptive, semantic segmentation, unbalanced data.**

I. Introduction

COMPLEX semantic segmentation scenes that often exist in the wide field of view (WFV) remote sensing images (RSIs), which covers a wide geographical area while maintaining high image resolution and rich spectral information [1], [2], [3], are usually acquired by spacecraft, aircraft, or ground-based camera equipment. It can be widely used in land-cover classification, resource management, environmental monitoring, disaster management, and urban planning [4].

Manuscript received 23 January 2024; revised 11 March 2024; accepted 7 April 2024. Date of publication 16 April 2024; date of current version 25 April 2024. This work was supported in part by the Beijing Natural Science Foundation under Grant 4232013, in part by the National Natural Science Foundation of China under Grant 62301044, and in part by the BIT Research and Innovation Promoting Project under Grant 2023YCXY027. *(Corresponding author: Mengmeng Zhang.)*

The authors are with the School of Information and Electronics and Beijing Key Laboratory of Fractional Signals and Systems, Beijing Institute of Technology, Beijing 100081, China (e-mail: 398551870@qq.com; mengmengzhang@bit.edu.cn; liwei089@ieee.org; gaoyunhao@bit.edu.cn; guiyuanyuan@bit.edu.cn; zyx829625@163.com).

Digital Object Identifier 10.1109/TGRS.2024.3388528

Semantic segmentation plays an important role in RSI processing tasks, which aims to assign each pixel of the input image to different classes [5], such as buildings, vegetation, water bodies, etc. Pixel-level segmentation requires methods to identify the objects and capture their shape. In recent years, deep neural networks (DNNs) have developed rapidly. Compared with traditional machine learning methods that require manual design of features, DNN has shown many advantages. It can cope with different geographic information and data sources. It can also adaptively extract detailed texture feature as well as context information from massive data, which allows the whole system to realize end-to-end learning [6]. Therefore, DNN has the ability to conduct efficient semantic segmentation and get precise results in RS areas. Among them, the proposal of fully convolutional networks (FCNs) marks a new stage in the task of semantic segmentation. FCN changes the classification layer of the traditional convolutional network from the fully connected layer to the full convolutional layer with a convolution kernel size of 1, so that the network can adapt to any size image input [7]. Due to the natural advantages of scale invariance, shift invariance, and position invariance [8], the FCN has led to a subsequent series of works and these works have achieved state-of-the-art performance in the semantic segmentation of natural scene images [9].

There has been plenty of researches about semantic segmentation task in RSIs using FCNs. Among them, RSIs have a conspicuous feature that often contains complex scenes. RS scenes often contain a heterogeneous landscape with objects in various sizes, shapes, and materials [10]. Due to the limited receptive fields of the convolution kernels, the network often cannot take objects at long distances into account. Moreover, the imbalance frequency of different classes will also lead the network to be biased toward high-frequency classes, which has a negative impact on the segmentation result.

Fully considering the contextual information and detailed information helps the segmentation networks have a better comprehension of input RS images. It has been recognized that the contextual information offers significant cues for RS image segmentation [11]. For example, cultivated land is often adjacent to human structures while roads often do not appear in the middle of large wetlands. In the process of the convolutional receptive field increasing from small to large, we can capture information from small to large scales, so that

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:27:50 UTC from IEEE Xplore. Restrictions apply.

the contextual information we obtain becomes more semantic and comprehensive from local incomplete parts. At the same time, keeping detailed information can better preserve texture and edge information, which plays an important role in segmenting objects with nuances, such as cultivated land and woodlands, as well as small objects. Moreover, using some external force to make the network more biased toward the low-frequency classes also plays a positive role. Therefore, massive research has been carried out around increasing the receptive fields, extracting and fusing multiscale information, and paying more attention to low-frequency samples. In this article, we use "low-frequency class" and "high-frequency class" to describe the class in which pixels appear less or more often, respectively.

The DeepLab series [12], [13], [14] models introduced atrous convolution and discussed the dilated rate to extract more comprehensive contextual information. By reasonably connecting convolution kernels with different dilated rates in series, the receptive field of the convolution operation can be expanded without increasing the parameter amount. Improvements were made in [7] and [15] based on dilated convolution to improve the recognition accuracy of small targets and large-scale variation of objects in RSIs. As for multiscale information, the shallow network extracts more textures and detailed information of images, while the deep network extracts more semantic information. The classic multiscale information fusion models are as follows. Pyramid scene parsing network (PSPNet) [16] uses pyramid pooling to aggregate feature information of different scales. An improved pyramid partial decoder (PPD) was proposed in [17] to make full use of encoder features and receptive field blocks. High-resolution network (HRNet) [18] focused on processing high-resolution images and achieved better performance by retaining multiscale information and performing feature fusion at multiple resolution levels. Cheng and Fu [19] and Seong and Choi [20], respectively, verified and improved HRNet by adding channels and spatial attention thresholds.

Another proven method that can extract and fuse feature from different scale efficiently is UNet [21]. It is a deep learning architecture designed for image segmentation, consisting of an encoder and a decoder that form a U-shaped network. The encoder progressively extracts image features, reducing spatial resolution, while the decoder, through upsampling operations, gradually restores the original resolution and fuses high-level features with low-level details. This allows the model to capture both global context and local details, facilitating accurate semantic segmentation. The skip connection combination method, as a feature-merging method, is really groundbreaking and classic, so many improvements are derived about block structures or combination process. U-NetFormer [22] proposed adding a vision transformer (ViT) mechanism to the decoder and combined it with convolutional neural network (CNN) to extract global and local information. Similarly, a proposed network GLOTS [23] built a global–local transformer segmentor by adopting transformers for both encoding and decoding, which enabled the network to acquire consistent feature and achieve good segmentation performance. The MA-U-Net [24] proposed by containing a multihead attention encoder was developed to enhance the network's ability to extract fine-grained features. Gui et al. [25] added an infrared band attention branch based on U-Net to improve the effect of the woodlands classes. In order to solve this problem of classes imbalance, the attempts made by researchers can generally be divided into two classes. One is the main improvement based on data, and the other is to reduce the number of high-frequency samples while increasing the number of low-frequency samples, such as undersampling high-frequency classes, resampling low-frequency classes, data enhancement, etc. [26]. Another idea is to make improvements in the training strategy. By setting the loss function, the learning process of the network is more biased toward small samples, such as increasing the weight of the low-frequency sample loss function [27], [28], or adding focus loss, dice loss [29], [30] on the basis of the cross-entropy loss function.

The above methods have achieved good results in the field of semantic segmentation of RSIs. However, there are still some problems.

1) It is often difficult for the above methods to consider both the lightweight of the network and the multiscale feature extraction (MS). Moreover, in the fusion of multiscale features, simple fusion methods such as direct splicing and addition are often adopted, without fully considering the relationship between features or paying attention to the unity of features at different scales at the same location.
2) Many previous studies mentioned above focused on how to improve the segmentation effect of low-frequency samples, but this may also lead to the cost of more high-frequency classes misclassification.

We pay attention to these problems and proposed a network which can segment different scale ground objects and deal with unbalanced classes in complex scenes. The method is as follows.

The scale-adaptive perception is integrated into the downsample process, which consists of two steps, feature extraction and feature fusion. For scale-adaptive feature extraction, we divide the features into different groups and extract information on various scales by using dilated convolution with different dilated rates. Depth-wise separable convolutions are also used to further cut down the number of parameters. For feature fusion step, features of diverse scales are fused through the scale-attention (SA) mechanism to reinforce the communication between scales and emphasize the information with high confidence. What is more, with setting up two segmentation heads for low-frequency samples and high-frequency samples, we used a weight map, which is strongly related to the sampling frequency, to reasonably incorporate the segmentation results.

According to above-mentioned methods, we have made the following contributions.

1) First, an MS method is brought up. It takes full advantages of the information along the channel dims and provides the network with diverse receptive fields. In this way, our method has the ability to obtain rich and varied information while the computation cost is kept as low as possible.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:27:50 UTC from IEEE Xplore. Restrictions apply.

2) Second, an SA mechanism is proposed. Based on self-attention mechanism, it weakens the information with large spatial difference caused by feature extraction at different scales. As a result, multiscale information is adaptively combined and strengthened with high spatial identity.
3) Third, a weight map strategy is presented additionally. With the aim of resolving the problem that the network's bias in the face of unbalanced classes, we create a separate branch to learn weight map, so that the accuracy of the high-frequency class and low-frequency class can be balanced adaptively.

The remainder of this article is organized as follows. The whole background and the related work are introduced in Section I. The proposed framework and strategy are depicted in Section II, respectively. The experimental results and analysis are presented in Section III. The ablation experiments on the innovation sites are described in Section IV. Finally, Section V summarizes with concluding remarks.

II. Methods

The whole structure of unbalanced class learning network with scale-adaptive perception (UCSANet) is displayed in Fig. 1. Feature maps extracted by the convolutional neural network have similar features with certain information redundancy in channel dimensions [31]. Based on that, the feature maps are split into different groups along the channel dimension to make full use of the redundancy and similar information between different bands of feature maps. Then, in order to adaptively acquire diverse scale feature, we set convolution with various dilated rate for each groups. At the same time, the depth-wise convolutional structure is used to extract space and channel features separately to further lightweight the network. In order to integrate information between different scales adaptively, we propose an SA mechanism, which fixes the spatial position and does self-attention calculation for each pixel. It makes full use of the feature information relationship between different scales in the same spatial position of the feature map. Moreover, to deal with unbalanced class, the segmentation head is redesigned. By setting different loss functions, the two segmentation heads biased toward large and small classes are set, respectively. Then, a weight map that has the same size as the input network image is introduced according to the classes' frequency, and the two segmentation results are adaptive combined to generate a final segmentation map based on weight map. As a result, the pixels of the low-frequency classes are biased toward the output results of the low-frequency segmentation head, and the pixels of the high-frequency classes are biased toward the high-frequency classes segmentation results.

A. Multiscale Feature Extraction

Dilated convolution, also known as atrous convolution, can increase the receptive field of the kernel without increasing the number of parameters or computational complexity. Different dilation rates determine different receptive fields, which can capture multiscale feature. However, because dilations create gaps between the kernel elements, they introduce a gridding effect leading to an uneven sampling of the input feature map and may bring a loss of fine-grained spatial information. Two techniques are taken to mitigate the gridding effect. First, as is shown in Table I, there are four branches in parallel, and different kernel sizes and dilated rates are chosen to balance between capturing global context and local details. Smaller dilation rates tend to maintain fine-grained spatial information, while larger rates capture larger contextual information. Second, each branch has two convolution operations sequentially, and the second one sets the dilated rate to 1. According to this, when the kernel size and dilated rate of multiple convolution layers in series meet certain conditions, the gap in the receptive field caused by dilated convolution can be filled.

TABLE I
Different Branch Convolution Settings and Corresponding Receptive Fields

Layer	conv	kernel	rate	Receptive fields
1	1	3	1	3×3
	2	3	1	
2	1	3	2	5×5
	2	3	1	
3	1	5	3	13×13
	2	5	1	
4	1	7	4	25×25
	2	7	1	

Meanwhile, depth-wise convolution with point-wise convolution (depth-wise separable convolutions) is designed for lightening parameters of the network while maintaining the origin capability of extraction features. Depth-wise convolution aims to capture the spatial information through channel-grouped convolution while point-wise convolution can capture the relationship between channels by convolution with 1×1 kernel size.

As illustrated in part I in Fig. 2, given a feature map $\mathbf{A} \in \mathbf{R}^{C\times H_f \times W_f}$, we assume that N kinds of scales are needed. Then we split it into N groups equally and ordinarily to get new feature $\mathbf{B} \in \mathbf{R}^{N\times(C/N)\times H\times W}$, where $\mathbf{B} = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_N\}$, $\mathbf{b}_i \in \mathbf{R}^{(C/N)\times H\times W}$. Next, we feed feature $\mathbf{B}$ into N convolution branches F, where $F = \{f_1, f_2, \ldots, f_N\}$, respectively. As is shown in part II in Fig. 2, branch f_i consists of two depth-wise convolution ${f_i}^{d1}$, ${f_i}^{d1}$ and one point-wise convolution ${f_i}^{p1}$ in series. In order to obtain different scale features from small to large, the dilated rate of ${f_i}^{d1}$ in different branches is gradually increased. Since the "gridding effect" will occur when conducting dilated convolution, the kernel size of sequential depth-wise convolution ${f_i}^{d1}$ should be set carefully to fill the gap in the receptive field. The information in the channel dimension in each group is then fused by ${f_i}^{p1}$. After that we can get feature $\mathbf{C} \in \mathbf{R}^{N\times(C/N)\times H\times W}$ that includes N different kinds of scales, where $\mathbf{C} = \{\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_N\}$, $\mathbf{c}_i \in \mathbf{R}^{(C/N)\times H\times W}$. The whole process can be written as

$$\begin{aligned} \mathbf{C} &= F(\mathbf{B}) \\ \mathbf{C} &= f_i(\mathbf{b_i}) = {f_i}^{p1}({f_i}^{d2}({f_i}^{d1}(\mathbf{b_i}))). \end{aligned} \tag{1}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:27:50 UTC from IEEE Xplore. Restrictions apply.

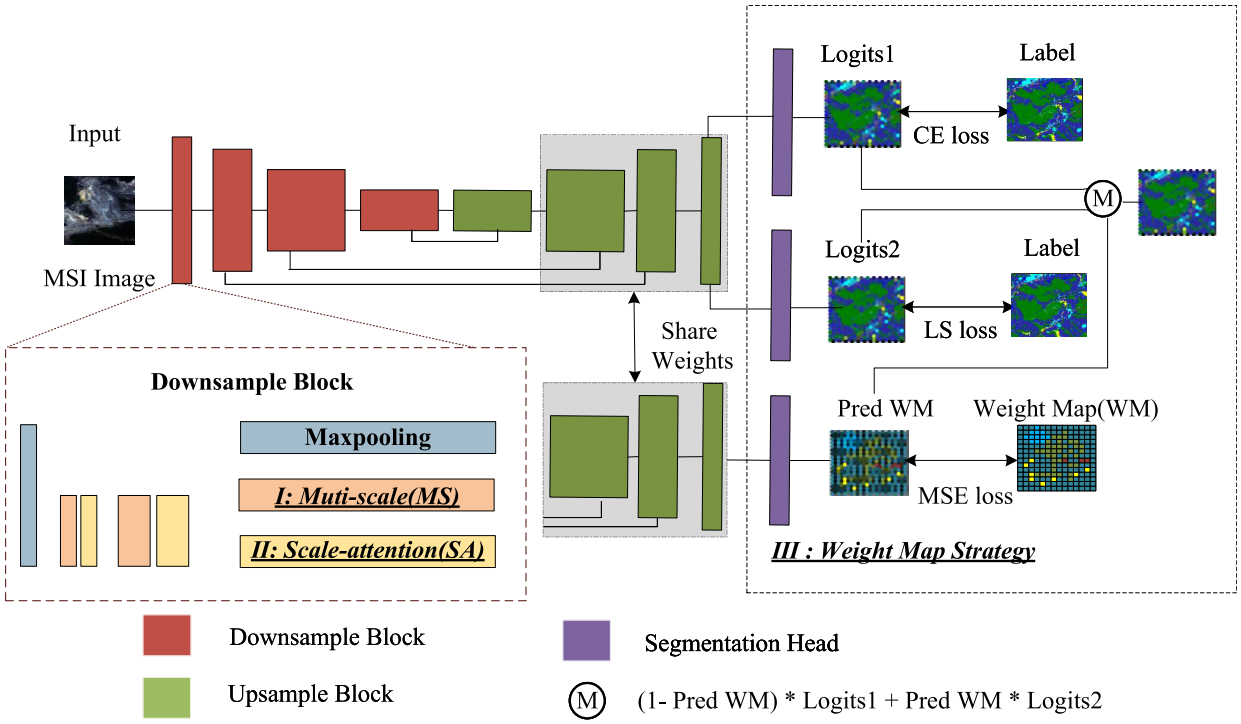


Fig. 1. Overall structure of the UCSANet for semantic segmentation with the detail of logits merging process through weight map. (Left) Diagrams of the entire structure and the downsample structure. (Right) Corresponding annotations.

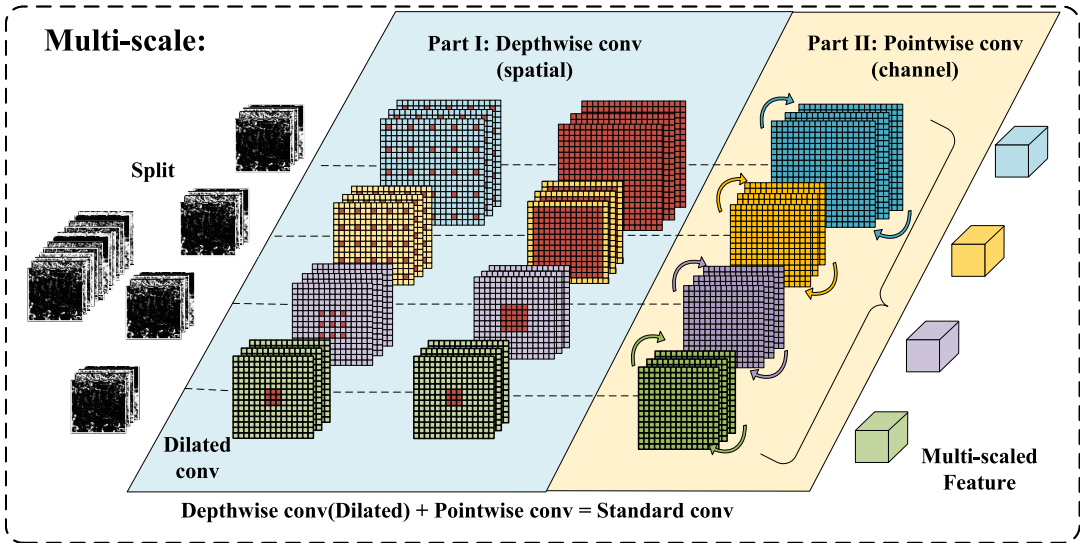


Fig. 2. Structure of MS. MS mainly includes two parts. Part I uses both depthwise convolution and dilated convolution to extract spatial feature of different receptive fields as well as to control the volume of parameters, while Part II uses pointwise convolution to extract useful features between channels.

B. Scale-Attention

Since different branches have different receptive fields, there may be some differences between the feature corresponding to pixels that have the same ground truth from different branches. These differences introduce intra-class inconsistency and affect the recognition accuracy [32]. To solve this problem, we build association among features from different branches with attention mechanisms to get more holistic information. This part can aggregate multiscale features adaptively and improve the consistency of different scale features at the same location.

As illustrated in Fig. 3, focusing on the dimension representing N scales and the one representing (C/N) channels feature, we transpose the feature $\mathbf{C} \in \mathbf{R}^{N\times(C/N)\times H\times W}$ from multiscale part and get $\mathbf{C}' \in \mathbf{R}^{(C/N)\times N\times H\times W}$. Then a matrix multiplication is performed between $\mathbf{C}'$ and $\mathbf{C}$, and a softmax layer is applied to calculate the SA map $\mathbf{S} \in \mathbf{R}^{N\times N\times H\times W}$, which means each pixel of the origin feature map that sizes $H \times W$ has an $N \times N$ attention map

$$\mathbf{D}_j{}^{(x,y)} = \sum_{i=1}^{N} \left(\mathbf{S}_{ji}{}^{(x,y)} \mathbf{C}_i{}^{(x,y)}\right) \tag{2}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:27:50 UTC from IEEE Xplore. Restrictions apply.

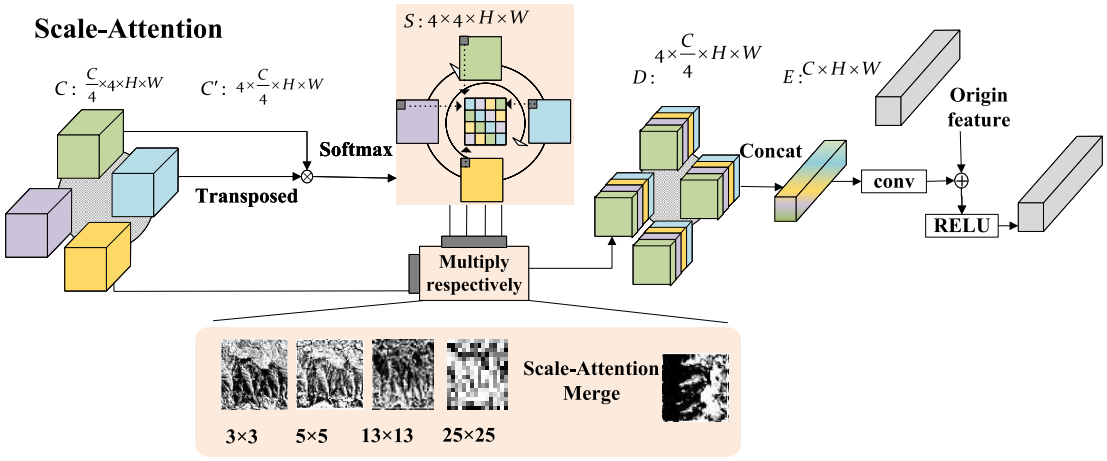


Fig. 3. Schematic of SA mechanism. We suppose that there are four groups ($N = 4$) and describe the specific meaning and shape of C, C', D, and E in this article.

where $\mathbf{D}$ is a weighted sum of features across all scales. Then we use a concatenation operation to fuse all reweighted scale features $\mathbf{E} \in \mathbf{R}^{C\times H\times W}$, which has the same size as the input feature A. To obtain the feature with required output channels $\mathbf{E}$, we convolve both $\mathbf{A}$ and $\mathbf{E}$ through pointwise convolution ${f_i}^{p2}$ and ${f_i}^{p2}$. Finally, we add them together to form a residual structure and feed it into rectified linear unit (ReLU) active function. As a result, we acquire the output feature which merges information adaptively from different scales without adding extra huge computation cost

$$\text{output} = \text{ReLU}({f_i}^{p2}(\mathbf{E}) + {f_i}^{p3}(\mathbf{A})). \tag{3}$$

C. Merging by Weight Map

Cross-entropy loss function is employed for the partial high-frequency segment head, while the Lovasz-Softmax is for the partial low-frequency segment head. A weight map, calculated based on the occurrence frequency of different classes, has the same size as the original image. The configuration principle of the weight map entails assigning a heavier weight to low-frequency classes and a lower weight to high-frequency classes. This enables adaptive selection of the final segmentation result by the network according to the weight map, favoring either:1) biasing pixels belonging to low-frequency classes toward the output result of the low-frequency segmentation head or 2) biasing pixels belonging to high-frequency classes toward their respective segmentation results.

1) Weight Map: The class frequency is calculated separately for each sample to build weight map. What is more, the value of all pixels for one class in weight map is the same. After getting a labeled image and its label $\mathbf{Label} \in \mathbf{R}^{H_{\text{ori}}\times W_{\text{ori}}}$, the frequency freq of all the classes of pixels is counted and recorded to generate a weight map $\mathbf{W}_{\text{gt}} \in \mathbf{R}^{H_{\text{ori}}\times W_{\text{ori}}}$ through these formulas

$$\begin{aligned} \text{freq}_i &= \frac{n_i}{\sum_i^N n_i} N \\ \text{weight}_i &= \frac{\log(\alpha)}{\log(\text{freq}_i + \alpha)} \\ \mathbf{W}_{\text{gt}}^{(x,y)} &= \text{weight}_i, \quad \text{if } \mathbf{Label}^{(x,y)} = i \end{aligned} \tag{4}$$

where N implies the total number of classes and represents ith class. freq_i stands for occurrence of ith class, and the lower freq_i is, the higher weight_i becomes. α is a value slightly bigger than 1 to keep $\text{weight}_i \in (0, 1)$. Then, according to $\mathbf{Label}^{(x,y)}$, we can get classes value i of the coordinate (x, y). Next we fill in the corresponding weight_i at the (x, y) position of $\mathbf{W}_{\text{gt}}$.

2) Merging Process: It is known that the cross-entropy loss is a logarithmic measure of the difference between the model's predicted probability and the actual label's probability distribution. However, for data sets with unbalanced classes, because the traditional cross-entropy function still treats each class equally, the classification effect of low-frequency classes is often poor. Fortunately, the Lovasz-Softmax loss function sorts errors between predictions and labels for different classes and then increases the weight of samples that are difficult to distinguish, which makes it a low-frequency sample-friendly loss function.

As illustrated in Fig. 1, the network architecture consists of an encoder part, a decoder part, and skip connections. The encoder part includes four downsample blocks, comprising two scale-adaptive blocks and a max-pooling layer with stride 2 for downsampling. Meanwhile, the decoder shares some features and ultimately has three loss functions to control the three optimization objectives. The later the new branch is separated, the more shared weights different branches possess. Consequently, the increase in the number of added parameters is minimized, leading to more similar predictions. Given that the primary objective of the segmentation head is to align predictions closely with the ground truth, fine-tuning a small percentage of the weights suffices to bias the prediction toward low or high-frequency categories, respectively. Conversely, the objective of another branch is to predict a weight map, which significantly deviates from the segmentation result. Hence, it is imperative to initiate weight learning earlier for this branch. Therefore, in light of computational efficiency and accuracy considerations, we structure our network accordingly. The main branch of the decoder develops two split headers at the last upsample block, using the traditional cross-entropy loss function and the Lovasz-Softmax loss function independently.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:27:50 UTC from IEEE Xplore. Restrictions apply.

Their outputs are $\mathbf{Logits}_{\text{Lovasz-Softmax}}$ and $\mathbf{Logits}_{\text{Cross-Entropy}}$, respectively, based on Label. For ease of narration, we abbreviate corner markers $_{\text{Lovasz-Softmax}}$ and $_{\text{Cross-Entropy}}$ to $_{\text{CE}}$ and $_{\text{LS}}$, respectively. There is another side-decoder derived from the second upsample block that shares the same structure as the main encoder but uses mean squared error (MSE) loss to learn from the weight map $\mathbf{W}_{\text{gt}}$ generated from Section III and outputs $\mathbf{W}_{\text{pred}}$. Ultimately, the outcome of the whole process can be calculated as follows:

$$\mathbf{Logits} = \mathbf{W}_{\text{pred}} * \mathbf{Logits}_{\text{LS}} + (1 - \mathbf{W}_{\text{pred}}) * \mathbf{Logits}_{\text{CE}} \quad (5)$$

which is designed to bias the end result of low-frequency classes toward $\mathbf{Logits}_{\text{LS}}$ through and the end result of high-frequency classes toward $\mathbf{Logits}_{\text{CE}}$.

III. Settings of Experiments and Discussion

A. Introduction of Dataset

The GF-6 satellite is a satellite equipped with multispectral RS instruments launched by the China National Space Administration (CSA), which is able to acquire WFV images with rich spectral information and high spatial resolution [33].

The RSI data we used is acquired by the GF-6 satellite, which has eight spectral bands from 0.4 to 0.89 μm. Data from eight cities in China, covering a total area of more than 163 000 km, spinning approximately 30° of latitude and 25° of longitude, is chosen to conduct our experiments. It includes Benxi, Chengde, Fangchenggang, Genhe, Huangshan, Puer, Tianshui, and Yichang provinces, respectively. As a matter of fact, the data comes from the Academy of Forestry, who are interested in forest land. The experts classify the objects contained in this dataset based on the national first-level classification system outlined in the Classification of Land Use Status, merge some artificial construction classes, and add wetland to meet demands of focusing on vegetation. The proportion of forest land is extremely large, which creates the dataset with unbalanced classes. Also, it contains scenes of different scales such as rivers, buildings, and farmland at the same time. Therefore, these characteristics make this dataset ideal for verifying the effectiveness of our proposed method.

In order to get labeled images, images sourced from the survey conducted by The State Forestry and Grassland Administration Survey Planning and Design Institute are sorted and made into vectors, and then marked with geographic information science (GIS) technology. After that, the WFV RSIs sourced from the satellite are processed through radiometric correction operation and aligned with survey images based on geographic coordinate information. As a result, the professionals divide data into eight classes in total, which contains wetland, cultivated land, plantation land, forests, grassland, roads and buildings, water resource, and others. The first seven classes are of our interests. As is shown in Fig. 4, forests account for the highest proportion, up to about 70%, followed by cultivated land, accounting for about 12%, and the remaining class is equal or less than 5%, which indicates that the occurrence frequency of different kinds of samples varies greatly.

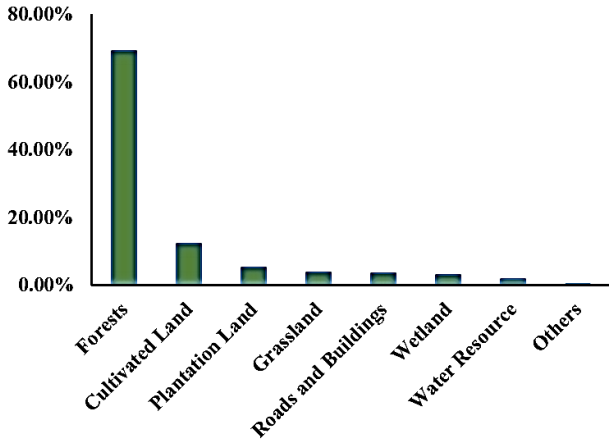


Fig. 4. Class frequency distribution. In descending order of frequency, the classes are forests, cultivated land, plantation land, grassland, roads and buildings, wetland, water resource, and others.

As is shown in Fig. 5, the target scales and the frequencies between different classes differ greatly in the complicated scene. For example, the water resource (red) and grassland (yellow) in the first picture, and roads and buildings (cyan) and cultivated land (purple) in the second picture differ greatly in the target scale. Also, it is clear that the frequency of forests (green) is much higher than that of other classes. All the images and the corresponding labels are split into 256 pixels × 256 pixels for convenience. After discarding them with clouds or misalignment, we finally select 3300 pairs of samples for training set, and 500 pairs of samples for verification and test set, respectively.

B. Settings of Experiments

The experiments are carried out on a computer with a GeForce RTX 3090 GPU with 24 GB of RAM. The optimizer function is the stochastic gradient descent (SGD) function. The learning rate starts at 0.01 and is reduced gradually by the ReduceLROnPlateau strategy. The output of the whole network structure is an eight-channel segmented prediction tensor. The eight channels of each pixel represent the probability the pixel might be in each of these eight classes. The training epoch is set to 150. The batch size number is set to 20. Both our method and comparison methods use the setup above.

C. Evaluation Metrics

In this article, overall accuracy (OA), and mean intersection over union (mIoU) are chosen to evaluate the overall segmentation, while precision, recall, and F1-score are chosen to analyze a single class. In particular, the F1-score can represent the overall segmentation effect of a certain type, while recall and precision can help us analyze the specific deviation of positive and negative samples. The calculation methods are

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:27:50 UTC from IEEE Xplore. Restrictions apply.

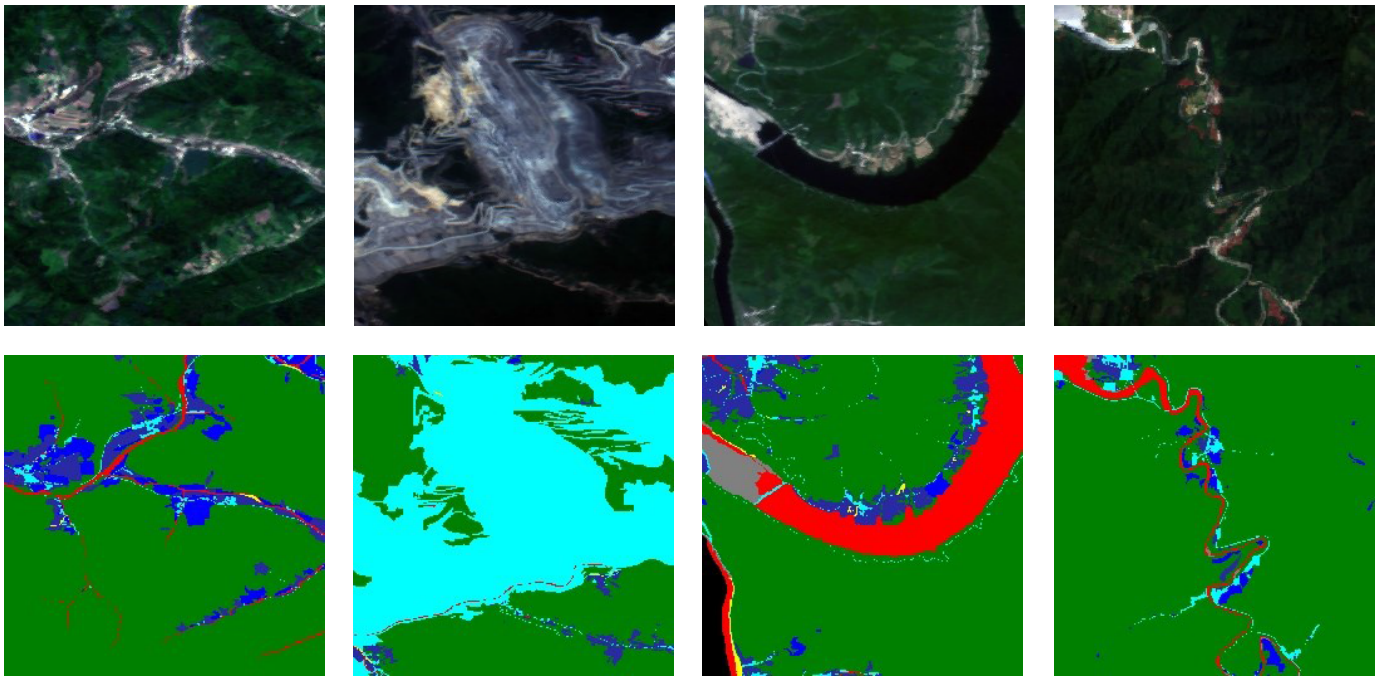


Fig. 5. Data sample, images in the first row are superimposed by the red band, green band, and blue band of the multispectral image taken by the GF-6 satellite, and images in the second row are their corresponding labels.

shown as follows:

$$\begin{cases} \text{Precision} = \dfrac{\text{TP}}{\text{TP}+\text{FP}} \\ \text{Recall} = \dfrac{\text{TP}}{\text{TP}+\text{FN}} \\ \text{F1-score} = \dfrac{2\text{TP}}{\text{FP}+2\text{TP}+\text{FN}} \\ \text{mIoU} = \dfrac{1}{k}\displaystyle\sum_{i=1}^{k}\dfrac{\text{TP}_i}{\text{TP}_i+\text{FN}_i+\text{FP}_i} \end{cases} \tag{6}$$

where true positive (TP), false positive (FP), true negative (TN), and false negative (FN) are concepts set for a single class. Suppose for class A, TP is the number of pixels that is labeled as A and is predicted as A. FP is the number of pixels that is labeled as A and is predicted as other classes. TN is the pixel number that is labeled as other classes and is predicted as other classes. FN is the pixel number that is labeled as other classes and is predicted as A. k implies the number of classes.

D. Segmentation Performance

Comparison experiments are designed and accomplished to test the performance of our method. To ensure the rigor of the experiment, MMSegmentation, an open source semantic segmentation toolbox,[1] is used to complete all the experiments including our method. Also, for fairness, none of the pre-training weights are loaded, and cross-entropy loss-the most classic loss function of semantic segmentation-is used uniformly by all networks. What is more, the final value is determined based on the average of five times execution, which guarantees the reliability of the results.

Using ResNet50 as the backbone, we set main and auxiliary segmentation heads, respectively. The testing results of segmentation networks are listed in Table II. We select the typical networks FCN [7], PSPNET for extraction multiscale features [16]. We also choose Deeplab-v3 network [13] using dilated convolution to expand receptive fields and its improved version Deeplab-v3 plus [14], then select a series of networks using different attention mechanisms criss-cross attention network (CCNet) [34], segmentation transformer (Segformer) [35], point-wise spatial attention network (PSANet) [36], and dual attention network (DANNet) [32]. Among them, excluding the network proposed by us, Deeplab-v3 plus, DANNet, and Deeplab-v3 rank among the top three in the OA index, all exceeding 82.6%, while Deeplab-v3 plus, PSPNet, and DANNet rank among the top three in mIoU index, all exceeding 47%. While FCN, segformer, CCNetd network performance is relatively not very good, as their mIoU is about equal to or less than 45%. In comparison, the proposed model UCSANet performs the most optimal result that scores 84.67% for OA and 52.27% for mIoU, in which OA is improved by about 1.5% and the mIoU is improved by about 4.4% compared with the best results of other comparison methods.

From the perspective of class, the segmentation effect of each class can be better judged by the IoU and F1-score of each class. As is shown in Table II, all of the classes reach the highest F1-score and IoU among all methods. F1-score is used here to specify, they are 73.69% for wetland, 71.13% for cultivated land, 51.49% for plantation land, 92.24% for forests, 45.61% for grassland, 61.23% for roads and buildings, and 7% for water resource. The IoU shares the same trend with F1-score.

[1]MMSegmentation Contributors, "OpenMMLab Semantic Segmentation Toolbox and Benchmark," Jul. 2020. [Online]. Available: https://github.com/open-mmlab/mmsegmentat.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:27:50 UTC from IEEE Xplore. Restrictions apply.

TABLE II
COMPARISON OF THE EVALUATION METRIC SCORES (%) AMONG DIFFERENT SEGMENTATION NETWORKS

Evaluation	Classes	FCN [7]	PSPNet [16]	Deeplab-v3 [13]	Deeplab-v3plus [14]	CCNet [34]	Segformer [35]	PSANet [36]	DANNet [32]	UCSANet
recall	Wetland	50.62	66.01	70.01	69.02	63.62	56.40	64.26	69.62	61.71
	Cultivated Land	69.36	66.68	70.37	73.48	70.65	73.59	67.65	70.89	76.48
	Plantation Land	35.41	43.50	39.64	35.89	35.79	27.53	40.74	32.22	47.72
	Forests	95.22	93.82	94.11	94.11	4.25	94.42	94.46	94.82	93.08
	Grassland	30.26	38.65	27.97	35.01	27.25	25.77	31.49	32.04	31.73
	Roads and Buildings	42.63	40.47	42.93	44.93	39.65	39.81	44.30	43.29	52.48
	Water Resource	53.47	61.25	54.26	57.94	53.44	54.59	53.25	56.08	60.10
IoU	Wetland	45.56	53.86	55.24	53.01	52.89	44.66	54.11	58.02	**58.35**
	Cultivated Land	51.46	50.64	51.30	53.94	50.36	51.35	50.61	51.84	**57.64**
	Plantation Land	28.39	30.61	30.20	28.60	27.15	22.65	30.29	26.51	**34.67**
	Forests	83.51	83.47	83.63	84.27	83.52	83.53	83.66	84.05	**85.60**
	Grassland	25.47	29.51	24.29	27.80	23.04	22.04	26.12	26.66	**29.55**
	Roads and Buildings	34.64	34.62	34.53	36.84	33.14	32.76	34.76	35.47	**44.12**
	Water Resource	47.99	48.37	47.85	50.58	47.26	48.46	47.38	48.83	**55.94**
F1-score	Wetland	62.60	70.01	71.04	69.29	69.19	61.75	70.22	73.43	**73.69**
	Cultivated Land	67.95	67.24	67.83	70.08	66.99	67.86	67.21	68.28	**73.13**
	Plantation Land	44.22	46.87	46.64	44.47	42.71	36.93	46.50	41.91	**51.49**
	Forests	91.02	90.99	91.05	91.46	91.02	91.02	91.10	91.34	**92.24**
	Grassland	40.61	45.57	39.89	43.50	37.45	36.12	41.43	42.10	**45.61**
	Roads and Buildings	51.45	51.43	51.40	53.84	49.78	49.36	51.59	52.36	**61.23**
	Water Resource	64.85	65.20	64.78	67.18	64.18	65.28	64.30	65.62	**71.74**
OA		82.56	82.45	82.63	83.18	82.24	82.05	82.57	83.02	**84.67**
mIoU		45.29	47.29	46.72	47.86	45.34	43.63	46.71	47.34	**52.27**

Fig. 6 shows a comparison figure of the segmentation results of five different methods and our method. The images from left to right are the labels, PSPNet, deeplabv3-plus, segformer, PSANet, DANNet, and our proposed UCSANet. It can be clearly seen that in response to complex scenes, our proposed algorithm can not only accurately identify ground objects, but also pay more attention to the details of ground objects. The phenomenon of missing and mis-marking in other segmentation methods is relatively more serious, and it is often impossible to accurately identify the drastic changes and subtle edges. However, our method alleviates this phenomenon by expanding the receptive field and MS, and the segmentation is more refined. At the same time, the addition of a weight map also improves the segmentation effect of small classes. In general, our proposed method is more targeted to complex RS semantic segmentation scenarios and has more advantages than other methods.

IV. ABLATION STUDY AND DISCUSSION

To verify the effectiveness of each block, this section progressively adds each structure to the base network and performs ablation experiments. UNet is selected as the base network.

A. Ablation Study

1) Ablation for Logits Merging: To explore the role of each structure in UCSANet with weight map strategy, we conduct another experiment and get the result in Table III. Also, for the sake of convenience, the abbreviation of each ablation study is illustrated in Table IV and is used in the following statement. Generally speaking, another two segmentation heads are added to the base network to conduct our weight map strategy. Scale-adaptive part can be easily embedded in any convolution network. The first four columns of experimental results are intended to illustrate the effects of the weight map. What is more, the metric scores in the fifth and the sixth column indicate the positive role the MS part and the SA part play, respectively. As a result, by comparing the first column and the last column, it is clear to see that the method proposed by us enhances the segmentation capability of the network from baseline. To be specific, OA improves for about 0.94% while mIoU improves dramatically for about 3.17%. Also, F1-score of seven classes improves by 0.27% for forests, 2.01% for cultivated land, 4.79% for plantation land, 6.72% for grassland, 3.25% for roads and buildings, 2.55% for wetland, 2.67% for water resource. Details of each experiment are as follows.

Table III shows the result of training U-Net with four kinds of loss, that is, "C," "L," "C + L," and "C-L-W," respectively, for the sake of analysis. Notice that the class names from top to bottom are arranged in order of frequency from most to least.

Comparing "C" and "L," we could tell that except for the two types of "forests" and "cultivated land" with higher frequencies, other types of F1 have been improved compared to cross-entropy loss, and mIoU is also improved, which means Lovasz-Softmax function can indeed make the model more inclined to low-frequency samples. However, The accuracy drops significantly, by about 2.17%. To analyze the reasons, we notice that for most low-frequency classes, the precision goes down and the recall goes up, which indicates that while the FN of this class decreases, more FP increases. High-frequency samples show the opposite trend simultaneously. According to this phenomenon, we could easily infer that more pixels are divided into low-frequency classes at the cost of the misclassification of high-frequency samples. Therefore, a greater percentage of errors leads to the overall segmentation effect decline.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:27:50 UTC from IEEE Xplore. Restrictions apply.

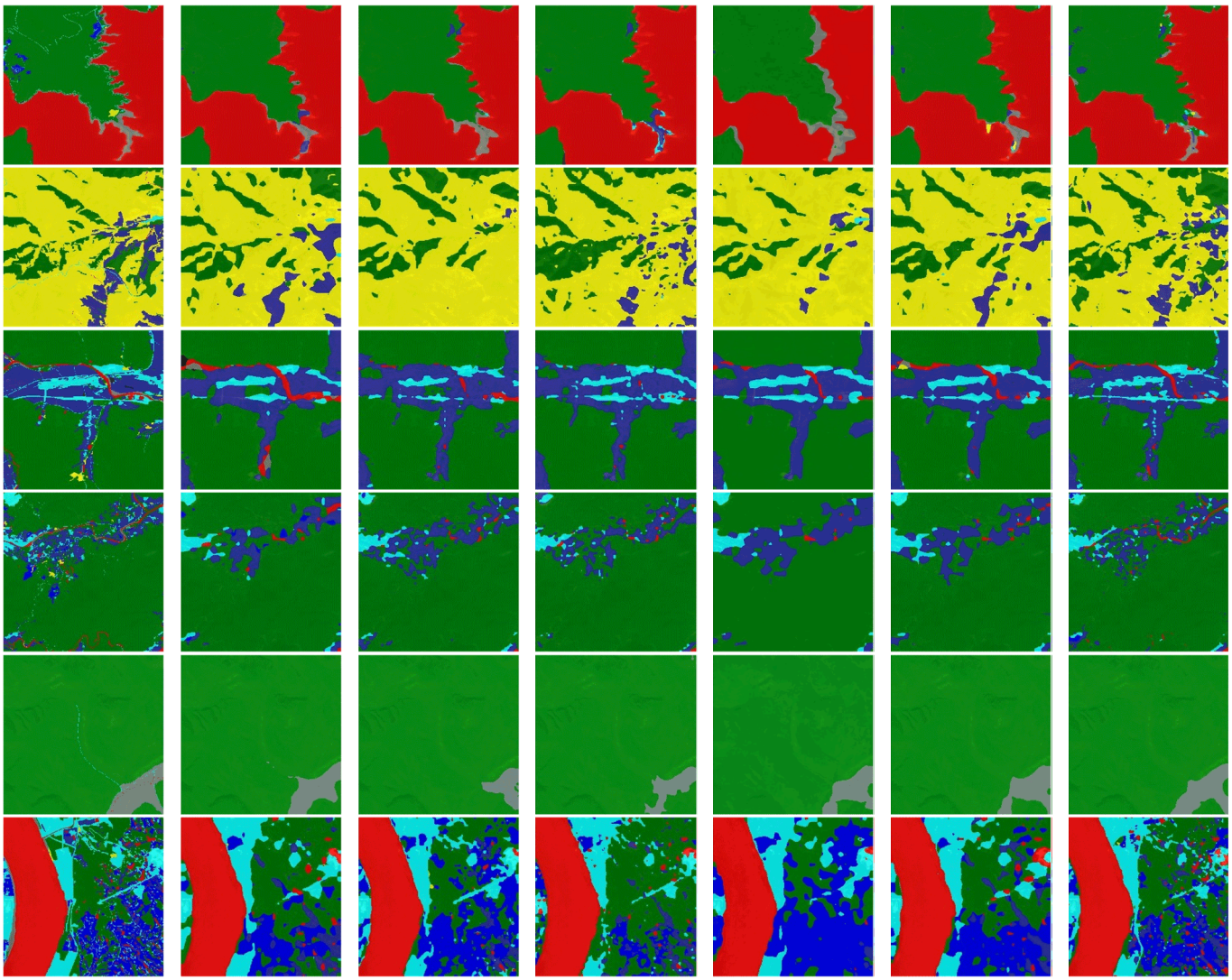


Fig. 6. Segmentation results by UCSANet and some other networks. These results show the comparison of the fineness of the segmentation by UCSANet and some other networks. (a) LABEL. (b) PSPNet. (c) DeeplabV3-plus. (d) Segformer. (e) PSANet. (f) DANNet. (g) UCSANet.

To solve this problem, first, as "C + L" shows, we simply add two kinds of loss functions together. Adding Lovasz-Softmax loss function directly to the baseline network improves the F1-score for some low-frequency classes, and mIoU improves by 0.29% as well. However, the OA decreases slightly, which suggests that adding Lovasz-Softmax loss function simply together is not an ideal strategy. Then, after we use the weight map to adaptively adjust the weight of the two loss functions ("C-L-W"), OA increases by 0.11% and mIoU increases by 0.86%. Importantly, in addition to the F1-score of forests being basically unchanged, the six other classes improves by 0.38% for cultivated land, 0.25% for plantation land, 1.94% for grassland, 1.69% for roads and buildings, 0.82% for wetland, and 1.04% for water resource.

2) Ablation for Multiscale Feature Extraction Part: To explore the role of the MS Part, we conduct another experiment "C-L-MS-W" and get the result in the fifth column of Table III. We replace the double-convolution block in UNet with our MS part. It is speculated that the MS part improves the segmentation capability by capturing multireceptive field information based on the comparison between the fourth and the fifth column of Table III. The result shows that OA improves for about 0.84% while mIoU improves for about 2.81%. Also, the F1-score of seven classes improves by 0.3% for forests, 1.95% for cultivated land, 5.06% for plantation land, 5.84% for grassland, 2.37% for both wetland and buildings, and 2.03% for water resource.

3) Ablation for Scale-Attention Part: On the basis of the above improvements, we further add an SA mechanism ("C-L-MSSA-W") to merge features of different scales better. The result is shown in the sixth column of Table III, which indicates that this mechanism can further improve the performance of the network. OA improves for about 0.1% and reaches 84.67% while mIoU improves for about 0.36% and reaches 52.27%. Moreover, there are five classes whose F1-score achieves the highest. They are 73.13% for cultivated land, 45.61% for grassland, 61.23% for roads and buildings, 73.69% for wetland, and 71.74% for water resource.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:27:50 UTC from IEEE Xplore. Restrictions apply.

TABLE III
SCORE (%) OF ABLATION STUDY OF UCSANET

CLASS \ PERCENTAGE (%)	EVALUATION	MODEL					
		Cross-Entropy	Lovazs-softmax	Cross-Entropy Lovazs-softmax (one-to-one add)	Cross-Entropy Lovazs-softmax Weightmap	Cross-Entropy Lovazs-softmax Weightmap Multi-scale feature	Cross-Entropy Lovazs-softmax Weightmap Multi-scale feature Scale-attention
Forests(69.2)	precision	89.79	92.18	89.93	89.72	90.47	90.16
	recall	94.26	89.73	93.87	94.19	94.14	94.42
	F1-score	91.97	90.94	91.86	91.90	**92.27**	92.24
Cultivated Land(12.3)	precision	67.85	67.84	67.86	69.06	69.80	71.44
	recall	74.71	72.49	75.05	74.12	76.67	74.91
	F1-score	71.12	70.09	71.27	71.50	73.07	**73.13**
Plantation Land(5.3)	precision	55.31	48.46	55.16	56.68	59.79	60.18
	recall	40.41	46.83	37.91	40.07	45.63	45.00
	F1-score	46.70	47.63	44.94	46.95	**51.76**	51.49
Grassland(3.8)	precision	54.54	30.47	50.11	51.98	54.85	54.25
	recall	30.22	57.71	32.24	33.62	37.76	39.35
	F1-score	38.89	39.88	39.23	40.83	44.73	**45.61**
Roads and Buildings(3.6)	precision	67.53	64.84	65.84	67.66	70.02	70.02
	recall	50.79	54.26	54.58	53.36	53.02	54.40
	F1-score	57.98	59.08	59.69	59.67	60.35	**61.23**
Wetland(3.2)	precision	74.07	78.28	74.80	73.41	75.93	76.28
	recall	68.44	65.60	68.09	70.57	71.25	71.28
	F1-score	71.14	71.38	71.28	71.96	73.51	**73.69**
Water Resource(1.8)	precision	78.65	82.96	78.94	82.22	81.57	81.51
	recall	61.57	61.60	63.25	61.10	63.01	64.07
	F1-score	69.07	70.71	70.23	70.11	71.10	**71.74**
OA		83.73	81.56	83.60	83.84	84.57	**84.67**
mIoU		49.10	49.37	49.39	49.96	51.91	**52.27**

TABLE IV
ABBREVIATION OF ABLATION STUDY

INDEX	ABBREVIASTION	FULL NAME
1	C	Cross-Entropy
2	L	Lovasz-Softmax
3	C+L	Cross-Entropy Lovazs-softmax (one-to-one add)
4	C-L-W	Cross-Entropy Lovazs-softmax WeightMap
5	C-L-MS-W	Cross-Entropy Lovazs-softmax WeightMap Multi-scale Feature
6	C-L-MSSA-W	Cross-Entropy Lovazs-softmax WeightMap Multi-scale Feature Scale-attention

TABLE V
SCORE (%) OF THE DISCUSSION ABOUT WEIGHT MAP IN THE PROPOSED UCSANET

weight of MSE Loss (λ)		10	20	25	30
OA		83.75	83.80	**83.84**	83.80
mIoU		49.42	49.56	**49.96**	49.45
F1-score	Forests	91.85	91.88	91.90	91.94
	Cultivated Land	71.21	71.42	71.50	71.38
	Plantation Land	44.95	44.74	46.95	45.45
	Grassland	38.33	39.50	40.83	38.22
	Roads and Buildings	59.27	58.47	59.67	59.41
	Wetland	72.48	72.64	71.96	72.10
	Water Resource	70.22	70.65	70.11	70.09

4) Visual Case: In Fig. 7, we present two examples of the segmentation performance using different methods, which serves as compelling evidence supporting our earlier analyses. The first column displays the ground truth for an image, consisting of six classes, forests (green), categorized as high-frequency features, while roads and buildings (cyan), plantation land (dark blue), grassland (yellow), cultivated land (purple), and water resource (red) are all considered as low-frequency classes.

The second column reveals that a substantial portion of the pixels corresponding to water resource or grassland is not accurately predicted. This observation highlights the limitations of the baseline method, as it struggles to capture low-frequency or smaller samples effectively.

The third and fourth columns depict the results by using the Lovasz-Softmax function and a combination of the Lovasz-Softmax function and the cross-entropy function with a 1:1 ratio. As we previously discussed, these approaches show improvement in the prediction of low-frequency classes. However, they also exhibit a drawback, which is the misclassification of forests' pixels as plantation land, roads and buildings, grassland, or something else. This suggests that the network prioritizes the accuracy of low-frequency classes at the expense of high-frequency ones.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:27:50 UTC from IEEE Xplore. Restrictions apply.

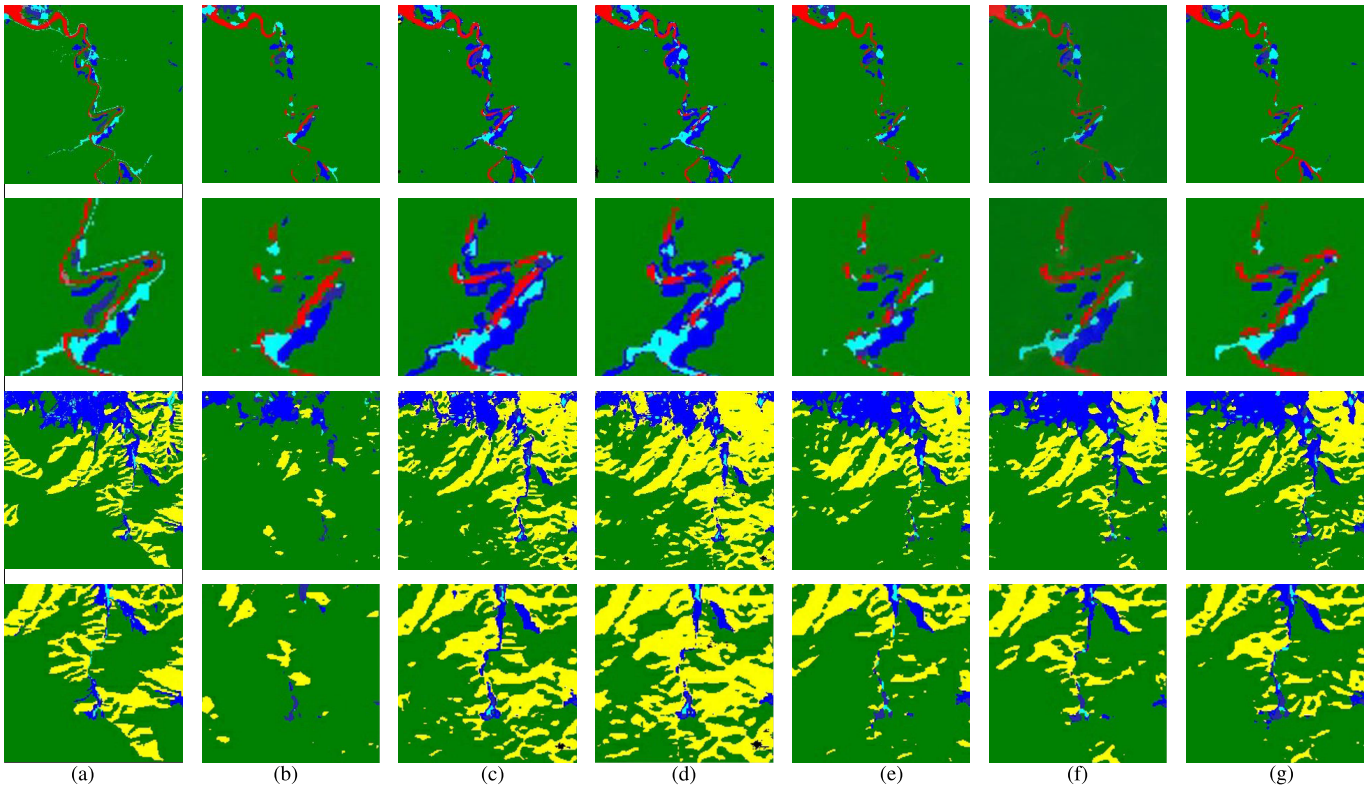


Fig. 7. Segmentation results of ablation studies of UCSANet. These results show the effect of each structure of our network. (a) LABEL. (b) C. (c) L. (d) C + L. (e) c-L-w. (f) c-L-MS-w. (g) c-L-UCSA-w.

In the fifth column, the inclusion of a weight map significantly addresses the aforementioned issue. It not only enhances the accuracy of predictions for low-frequency classes but also ensures the preservation of high-frequency forests classes. This result underscores the positive impact of introducing a weight map and merging two types of logits.

The improved performance showcased in the sixth column and seventh empowers the network with the ability to extract Scale-adaptive features and employ SA as an effective feature fusion method. Consequently, this further enhances the performance of object segmentation, which is the closest prediction to the ground truth. We are increasingly confident in the importance of Scale-adaptive information for accurately segmenting objects and have validated the effectiveness of our methodology.

B. Discussion

1) Discussion About the Loss Function: Since we design the main branch with two different loss functions Loss_{LS} and Loss_{CE} learning from segmentation label and the side branch with one loss function MSELoss learning from weight map, the total loss function Loss is as follows:

$$\text{Loss} = \text{Loss}_{\text{CE}} + \text{Loss}_{\text{LS}} + \lambda \text{Loss}_{\text{MSE}} \tag{7}$$

where λ is a hyper-parameter that can control the proportion of importance of learning weight map in the multitask experiment. In order to see the effect of the weight map more clearly, we change λ. Inspired by Kendall et al. [37], we try to consider the homoscedastic uncertainty between each task to determine the approximate range of weights for two task loss functions, and then get the result shown in Table V. With the increasing of the weight of MSE loss, mIoU goes up first and then goes down. The analysis of underlying factors suggests that when the value of lambda is excessively small, the prediction of the weight map is not accurate enough to balance the predictions of the two loss functions. Meanwhile, when λ is too large, the model might ignore the importance of learning to segment the results accurately. This part of the discussion further proves the effectiveness of the weight map and implies the requirements of selecting an appropriate weight. We choose 25 as the weight in the above experiments.

V. Conclusion

In this article, UCSANet, a scale-adaptive network with weight map strategy is proposed. The scale-adaptive perception part is composed of MS part and SA for feature merging. In the MS part, features are divided into four groups to make full use of the duplicate information in the channel dimension at first. Then, depth-wise and point-wise convolutions combined with different dilated rates are used to extract scale-adaptive features in different groups. After extracting scale-adaptive features, the self-attention mechanism is modified at the scale level to merge features more intelligently and efficiently. Moreover, a weight map strategy is used to deal with complex RS scenes and balance the learning of high-frequency and low-frequency classes. Experiments are conducted to verify how each part improves the performance of network segmentation. Finally, in a comprehensive experiment of segmenting WFV RSIs acquired by GF-6 satellite,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:27:50 UTC from IEEE Xplore. Restrictions apply.

the model is proved to have better performance compared to other deep learning-based semantic segmentation methods.

REFERENCES

[1] S. Mei, Y. Geng, J. Hou, and Q. Du, "Learning hyperspectral images from RGB images via a coarse-to-fine CNN," *Sci. China Inf. Sci.*, vol. 65, no. 5, pp. 1–14, May 2022.

[2] M. Zhang, W. Li, X. Zhao, H. Liu, R. Tao, and Q. Du, "Morphological transformation and spatial-logical aggregation for tree species classification using hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501212.

[3] Y. Gao et al., "Hyperspectral and multispectral classification for coastal wetland using depthwise feature interaction network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512615.

[4] M. Mehmood, A. Shahzad, B. Zafar, A. Shabbir, and N. Ali, "Remote sensing image classification: A comprehensive review and applications," *Math. Problems Eng.*, vol. 2022, pp. 1–24, Aug. 2022.

[5] Z. Wang, J. Guo, W. Huang, and S. Zhang, "High-resolution remote sensing image semantic segmentation based on a deep feature aggregation network," *Meas. Sci. Technol.*, vol. 32, no. 9, May 2021, Art. no. 095002.

[6] T. Hoeser, F. Bachofer, and C. Kuenzer, "Object detection and image segmentation with deep learning on Earth observation data: A review—Part II: Applications," *Remote Sens.*, vol. 12, no. 18, p. 3053, Sep. 2020. [Online]. Available: https://www.mdpi.com/2072-4292/12/18/3053

[7] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," 2014, *arXiv:1411.4038*.

[8] S. Mei, R. Jiang, M. Ma, and C. Song, "Rotation-invariant feature learning via convolutional neural network with cyclic polar coordinates convolutional layer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5600713.

[9] G. Chen et al., "SDFCNv2: An improved FCN framework for remote sensing images semantic segmentation," *Remote Sens.*, vol. 13, no. 23, p. 4902, Dec. 2021.

[10] R. Liu, L. Mi, and Z. Chen, "AFNet: Adaptive fusion network for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7871–7886, Sep. 2021.

[11] L. Mou, Y. Hua, and X. X. Zhu, "Relation matters: Relational context-aware fully convolutional network for semantic segmentation of high-resolution aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7557–7569, Nov. 2020.

[12] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "Semantic image segmentation with deep convolutional nets and fully connected CRFs," 2014, *arXiv:1412.7062*.

[13] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," Dec. 2017, *arXiv:1706.05587*.

[14] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.

[15] Z. Li et al., "Cascaded multiscale structure with self-smoothing Atrous convolution for semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jun. 2022, Art. no. 5605713.

[16] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2881–2890.

[17] H. Wei, X. Xu, N. Ou, X. Zhang, and Y. Dai, "DEANet: Dual encoder with attention network for semantic segmentation of remote sensing imagery," *Remote Sens.*, vol. 13, no. 19, p. 3900, Sep. 2021.

[18] J. Wang et al., "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, Oct. 2020.

[19] Z. Cheng and D. Fu, "Remote sensing image segmentation method based on HRNET," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Sep. 2020, pp. 6750–6753.

[20] S. Seong and J. Choi, "Semantic segmentation of urban buildings using a high-resolution network (HRNet) with channel and spatial attention gates," *Remote Sens.*, vol. 13, no. 16, p. 3087, Aug. 2021.

[21] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention—MICCAI*, Munich, Germany. Springer, 2015, pp. 234–241.

[22] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022.

[23] Y. Liu, Y. Zhang, Y. Wang, and S. Mei, "Rethinking transformers for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5617515.

[24] Y. Sun, F. Bi, Y. Gao, L. Chen, and S. Feng, "A multi-attention UNet for semantic segmentation in remote sensing images," *Symmetry*, vol. 14, no. 5, p. 906, Apr. 2022.

[25] Y. Gui, W. Li, X.-G. Xia, R. Tao, and A. Yue, "Infrared attention network for woodland segmentation using multispectral satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5627214.

[26] Z. Zhou et al., "A dynamic effective class balanced approach for remote sensing imagery semantic segmentation of imbalanced data," *Remote Sens.*, vol. 15, no. 7, p. 1768, 2023.

[27] A. Zheng, J. He, M. Wang, C. Li, and B. Luo, "Category-wise fusion and enhancement learning for multimodal remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4416212.

[28] X. Yang, X. Fan, M. Peng, Q. Guan, and L. Tang, "Semantic segmentation for remote sensing images based on an AD-HRNet model," *Int. J. Digit. Earth*, vol. 15, no. 1, pp. 2376–2399, Dec. 2022.

[29] Y. Liu, S. Mei, S. Zhang, Y. Wang, M. He, and Q. Du, "Semantic segmentation of high-resolution remote sensing images using an improved transformer," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 3496–3499.

[30] M. Wang, H. Du, S. Xu, and G. N. Surname, "Remote sensing image segmentation of ground objects based on improved Deeplabv3+," in *Proc. IEEE Int. Conf. Ind. Technol. (ICIT)*, Aug. 2022, pp. 1–6.

[31] K. Han, Y. Wang, Q. Tian, J. Guo, C. Xu, and C. Xu, "GhostNet: More features from cheap operations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 1580–1589.

[32] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.

[33] S. Xu et al., "Remote sensing of chlorophyll-a in Xinkai Lake using machine learning and GF-6 WFV images," *Remote Sens.*, vol. 14, no. 20, p. 5136, Oct. 2022.

[34] Z. Huang et al., "CCNet: Criss-cross attention for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 6, pp. 6896–6908, Jun. 2023.

[35] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, "SegFormer: Simple and efficient design for semantic segmentation with transformers," 2021, *arXiv:2105.15203*.

[36] H. Zhao et al., "Psanet: Point-wise spatial attention network for scene parsing," in *Proc. ECCV*, 2018, pp. 267–283.

[37] R. Cipolla, Y. Gal, and A. Kendall, "Multi-task learning using uncertainty to weigh losses for scene geometry and semantics," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 7482–7491.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:27:50 UTC from IEEE Xplore. Restrictions apply.